

CIS41032 - PRACTICAL FOR ADVANCE DATABASE MANAGEMENT SYSTEM

CA 01

Date:2026/06/02

Reg.No:SEU/IS/20/ICT/084

Index.No:ICT461

01.

use HospitalDB

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
Microsoft Windows [Version 10.0.19045.6466]
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C:\Users\Kalhara>mongosh
Current Mongosh Log ID: 6a1e9dd5e5f996e53a3682d0
Connecting to:      mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.8.2
Using MongoDB:      8.0.3
Using Mongosh:       2.8.2
mongosh 2.8.3 is available for download: https://www.mongodb.com/try/download/shell

For mongosh info see: https://www.mongodb.com/docs/mongosh-shell/

-----
The server generated these startup warnings when booting
  2026-06-01T23:54:56.978+05:30: Access control is not enabled for the database. Read and write access to data and configuration is unrestricted
-----

test> // 1. Create Database

test> use HospitalDB
switched to db HospitalDB
HospitalDB>
```

02.

```
db.patients.insertMany([
  {patient_id:"P001",name:"Ravi
Kumar",age:45,gender:"Male",disease:"Diabetes",ward:"A1",admission_date:"2023-09-
12"},
  {patient_id:"P002",name:"Saman
Perera",age:30,gender:"Male",disease:"Fever",ward:"B2",admission_date:"2023-10-
05"},
  {patient_id:"P003",name:"Anjali
Deva",age:28,gender:"Female",disease:"Dengue",ward:"A3",admission_date:"2023-11-
01"},
  {patient_id:"P004",name:"Kumara Silva",age:56,gender:"Male",disease:"Heart
Disease",ward:"C1",admission_date:"2023-08-15"},
  {patient_id:"P005",name:"Nadeesha
Sen",age:35,gender:"Female",disease:"Asthma",ward:"A2",admission_date:"2023-12-
18"},
```

```
{patient_id:"P006",name:"Ishara  
Wick",age:60,gender:"Female",disease:"Arthritis",ward:"D1",admission_date:"2024-  
01-10"},  
{patient_id:"P007",name:"Akalanka  
Jay",age:40,gender:"Male",disease:"Hypertension",ward:"A3",admission_date:"2023-  
12-01"},  
{patient_id:"P008",name:"Ruwani  
Lak",age:32,gender:"Female",disease:"Diabetes",ward:"B1",admission_date:"2023-07-  
20"},  
{patient_id:"P009",name:"Mahesh  
Mad",age:25,gender:"Male",disease:"Fracture",ward:"C2",admission_date:"2023-09-  
08"},  
{patient_id:"P010",name:"Dinuli  
Thar",age:29,gender:"Female",disease:"Migraine",ward:"A1",admission_date:"2023-  
11-22"},  
{patient_id:"P011",name:"Suresh  
Band",age:38,gender:"Male",disease:"Fever",ward:"B2",admission_date:"2023-09-  
30"},  
{patient_id:"P012",name:"Chathura  
Ga",age:47,gender:"Male",disease:"Diabetes",ward:"A2",admission_date:"2023-10-  
18"},  
{patient_id:"P013",name:"Nimali  
Thil",age:34,gender:"Female",disease:"Asthma",ward:"A2",admission_date:"2024-01-  
01"},  
{patient_id:"P014",name:"Thilina  
Jay",age:31,gender:"Male",disease:"Dengue",ward:"B3",admission_date:"2023-11-  
15"},  
{patient_id:"P015",name:"Malee Perer",age:50,gender:"Female",disease:"Heart  
Disease",ward:"C1",admission_date:"2023-09-25"}  
])
```

❏ mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000

```
HospitalDB> db.patients.insertMany([
  {patient_id:"P001",name:"Ravi Kumar",age:45,gender:"Male",disease:"Diabetes",ward:"A1",admission_date:"2023-09-12"},
  {patient_id:"P002",name:"Saman Perera",age:30,gender:"Male",disease:"Fever",ward:"B2",admission_date:"2023-10-05"},
  {patient_id:"P003",name:"Anjali Deva",age:28,gender:"Female",disease:"Dengue",ward:"A3",admission_date:"2023-11-01"},
  {patient_id:"P004",name:"Kumara Silva",age:56,gender:"Male",disease:"Heart Disease",ward:"C1",admission_date:"2023-08-15"},
  {patient_id:"P005",name:"Nadeesha Sen",age:35,gender:"Female",disease:"Asthma",ward:"A2",admission_date:"2023-12-18"},
  {patient_id:"P006",name:"Ishara Wick",age:60,gender:"Female",disease:"Arthritis",ward:"D1",admission_date:"2024-01-10"},
  {patient_id:"P007",name:"Akalanka Jay",age:40,gender:"Male",disease:"Hypertension",ward:"A3",admission_date:"2023-12-01"},
  {patient_id:"P008",name:"Ruwan Lak",age:32,gender:"Female",disease:"Diabetes",ward:"B1",admission_date:"2023-07-20"},
  {patient_id:"P009",name:"Mahesh Mad",age:25,gender:"Male",disease:"Fracture",ward:"C2",admission_date:"2023-09-08"},
  {patient_id:"P010",name:"Dinuli Than",age:29,gender:"Female",disease:"Migraine",ward:"A1",admission_date:"2023-11-22"},
  {patient_id:"P011",name:"Suresh Band",age:38,gender:"Male",disease:"Fever",ward:"B2",admission_date:"2023-09-30"},
  {patient_id:"P012",name:"Chathura Ga",age:47,gender:"Male",disease:"Diabetes",ward:"A2",admission_date:"2023-10-18"},
  {patient_id:"P013",name:"Nimali Thil",age:34,gender:"Female",disease:"Asthma",ward:"A2",admission_date:"2024-01-01"},
  {patient_id:"P014",name:"Thilina Jay",age:31,gender:"Male",disease:"Dengue",ward:"B3",admission_date:"2023-11-15"},
  {patient_id:"P015",name:"Malee Perer",age:50,gender:"Female",disease:"Heart Disease",ward:"C1",admission_date:"2023-09-25"}
])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a1ea252e5f996e53a3682d1'),
    '1': ObjectId('6a1ea252e5f996e53a3682d2'),
    '2': ObjectId('6a1ea252e5f996e53a3682d3'),
    '3': ObjectId('6a1ea252e5f996e53a3682d4'),
    '4': ObjectId('6a1ea252e5f996e53a3682d5'),
    '5': ObjectId('6a1ea252e5f996e53a3682d6'),
    '6': ObjectId('6a1ea252e5f996e53a3682d7'),
    '7': ObjectId('6a1ea252e5f996e53a3682d8'),
    '8': ObjectId('6a1ea252e5f996e53a3682d9'),
    '9': ObjectId('6a1ea252e5f996e53a3682da'),
    '10': ObjectId('6a1ea252e5f996e53a3682db'),
    '11': ObjectId('6a1ea252e5f996e53a3682dc'),
    '12': ObjectId('6a1ea252e5f996e53a3682dd'),
    '13': ObjectId('6a1ea252e5f996e53a3682de'),
    '14': ObjectId('6a1ea252e5f996e53a3682df')
  }
}
HospitalDB>
```

03.

```
db.patients.insertOne({
  patient_id:"P016",
  name:"Kavindi Nirosh",
  age:37,
  gender:"Female",
  disease:"Fracture",
  ward:"C2",
  admission_date:"2024-02-14"
})
```

 mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000

```
HospitalDB> db.patients.insertOne({
|   patient_id:"P016",
|   name:"Kavindi Nirosh",
|   age:37,
|   gender:"Female",
|   disease:"Fracture",
|   ward:"C2",
|   admission_date:"2024-02-14"
| })
{
  acknowledged: true,
  insertedId: ObjectId('6a1ea27ce5f996e53a3682e0')
}
HospitalDB>
```

04.

```
db.patients.find({ward:"A2"})
```

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
HospitalDB> db.patients.find({ward:"A2"})
[
  {
    _id: ObjectId('6a1ea252e5f996e53a3682d5'),
    patient_id: 'P005',
    name: 'Nadeesha Sen',
    age: 35,
    gender: 'Female',
    disease: 'Asthma',
    ward: 'A2',
    admission_date: '2023-12-18'
  },
  {
    _id: ObjectId('6a1ea252e5f996e53a3682dc'),
    patient_id: 'P012',
    name: 'Chathura Ga',
    age: 47,
    gender: 'Male',
    disease: 'Diabetes',
    ward: 'A2',
    admission_date: '2023-10-18'
  },
  {
    _id: ObjectId('6a1ea252e5f996e53a3682dd'),
    patient_id: 'P013',
    name: 'Nimali Thil',
    age: 34,
    gender: 'Female',
    disease: 'Asthma',
    ward: 'A2',
    admission_date: '2024-01-01'
  }
]
HospitalDB>
```

05.

```
db.patients.find({disease:"Diabetes"})
```

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
HospitalDB> db.patients.find({disease:"Diabetes"})
[
  {
    _id: ObjectId('6a1ea252e5f996e53a3682d1'),
    patient_id: 'P001',
    name: 'Ravi Kumar',
    age: 45,
    gender: 'Male',
    disease: 'Diabetes',
    ward: 'A1',
    admission_date: '2023-09-12'
  },
  {
    _id: ObjectId('6a1ea252e5f996e53a3682d8'),
    patient_id: 'P008',
    name: 'Ruwan Lak',
    age: 32,
    gender: 'Female',
    disease: 'Diabetes',
    ward: 'B1',
    admission_date: '2023-07-20'
  },
  {
    _id: ObjectId('6a1ea252e5f996e53a3682dc'),
    patient_id: 'P012',
    name: 'Chathura Ga',
    age: 47,
    gender: 'Male',
    disease: 'Diabetes',
    ward: 'A2',
    admission_date: '2023-10-18'
  }
]
HospitalDB>
```

```
db.patients.find({gender:"Female"},{name:1,disease:1,_id:0})
```

07.

```
db.patients.find({admission_date:{$gt:"2023-11-01"}})
```

```

mongosh mongodbs://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
Hospital08> db.patients.find({admission_date:{$gt:"2023-11-01"}})
[
  {
    _id: ObjectId('6a1ea252e5f996e53a3682d5'),
    patient_id: 'P005',
    name: 'Nadeesha Sen',
    age: 35,
    gender: 'Female',
    disease: 'Asthma',
    ward: 'A2',
    admission_date: '2023-12-18'
  },
  {
    _id: ObjectId('6a1ea252e5f996e53a3682d6'),
    patient_id: 'P006',
    name: 'Tshara Wick',
    age: 60,
    gender: 'Female',
    disease: 'Arthritis',
    ward: 'D1',
    admission_date: '2024-01-10'
  },
  {
    _id: ObjectId('6a1ea252e5f996e53a3682d7'),
    patient_id: 'P007',
    name: 'Akaianka Jay',
    age: 40,
    gender: 'Male',
    disease: 'Hypertension',
    ward: 'A3',
    admission_date: '2023-12-01'
  },
  {
    _id: ObjectId('6a1ea252e5f996e53a3682da'),
    patient_id: 'P010',
    name: 'Dinuli Than',
    age: 29,
    gender: 'Female',
    disease: 'Migraine',
    ward: 'A1',
    admission_date: '2023-11-22'
  },
  {
  }
]

```

08.

```
db.patients.find().sort({age:-1})
```

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
HospitalDB> db.patients.find().sort({age:-1})
[
  {
    _id: ObjectId('6a1ea252e5f996e53a3682d6'),
    patient_id: 'P006',
    name: 'Ishara Wick',
    age: 60,
    gender: 'Female',
    disease: 'Arthritis',
    ward: 'D1',
    admission_date: '2024-01-10'
  },
  {
    _id: ObjectId('6a1ea252e5f996e53a3682d4'),
    patient_id: 'P004',
    name: 'Kumara Silva',
    age: 56,
    gender: 'Male',
    disease: 'Heart Disease',
    ward: 'C1',
    admission_date: '2023-08-15'
  },
  {
    _id: ObjectId('6a1ea252e5f996e53a3682df'),
    patient_id: 'P015',
    name: 'Mailee Perera',
    age: 50,
    gender: 'Female',
    disease: 'Heart Disease',
    ward: 'C1',
    admission_date: '2023-09-25'
  },
  {
    _id: ObjectId('6a1ea252e5f996e53a3682dc'),
    patient_id: 'P012',
    name: 'Chathura Ga',
    age: 47,
    gender: 'Male',
    disease: 'Diabetes',
    ward: 'A2',
    admission_date: '2023-10-18'
  }
]
```

09.

```
db.patients.find({gender:"Male",age:{$lt:40}})
```

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
HospitalDB> db.patients.find({gender:"Male",age:{$lt:40}})
[
  {
    _id: ObjectId('6a1ea252e5f996e53a3682d2'),
    patient_id: 'P002',
    name: 'Saman Perera',
    age: 30,
    gender: 'Male',
    disease: 'Fever',
    ward: 'B2',
    admission_date: '2023-10-05'
  },
  {
    _id: ObjectId('6a1ea252e5f996e53a3682d9'),
    patient_id: 'P009',
    name: 'Mahesh Mad',
    age: 25,
    gender: 'Male',
    disease: 'Fracture',
    ward: 'C2',
    admission_date: '2023-09-08'
  },
  {
    _id: ObjectId('6a1ea252e5f996e53a3682db'),
    patient_id: 'P011',
    name: 'Suresh Band',
    age: 38,
    gender: 'Male',
    disease: 'Fever',
    ward: 'B2',
    admission_date: '2023-09-30'
  },
  {
    _id: ObjectId('6a1ea252e5f996e53a3682de'),
    patient_id: 'P014',
    name: 'Thilina Jay',
    age: 31,
    gender: 'Male',
    disease: 'Dengue',
    ward: 'B3',
    admission_date: '2023-11-15'
  }
]
HospitalDB>
```

10.

```
db.patients.updateOne({patient_id:"P003"},{$set:{ward:"B1"}})
```

 mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000

```
HospitalDB> db.patients.updateOne({patient_id:"P003"},{$set:{ward:"B1"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
HospitalDB>
```

11.

```
db.patients.updateOne({patient_id:"P002"},{$set:{disease:"Viral Fever"}})
```

 mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000

```
HospitalDB> db.patients.updateOne({patient_id:"P002"},{$set:{disease:"Viral Fever"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
HospitalDB>
```


12.

```
db.patients.updateMany({admission_date:{$lt:"2023-09-01"}},{ $set:{status:"Discharged"}})
```

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
HospitalDB> db.patients.updateMany({admission_date:{$lt:"2023-09-01"}},{ $set:{status:"Discharged"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
HospitalDB>
```

13.

```
db.patients.updateMany({disease:"Heart Disease"},{ $set:{status:"Critical"}})
```

```
mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000
HospitalDB> db.patients.updateMany({disease:"Heart Disease"},{ $set:{status:"Critical"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
HospitalDB>
```

View All Records

```
db.patients.find().pretty()
```

❏ mongosh mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000

HospitalDB> db.patients.find().pretty()

```
[
  {
    _id: ObjectId('6a1ea252e5f996e53a3682d1'),
    patient_id: 'P001',
    name: 'Ravi Kumar',
    age: 45,
    gender: 'Male',
    disease: 'Diabetes',
    ward: 'A1',
    admission_date: '2023-09-12'
  },
  {
    _id: ObjectId('6a1ea252e5f996e53a3682d2'),
    patient_id: 'P002',
    name: 'Saman Perera',
    age: 30,
    gender: 'Male',
    disease: 'Viral Fever',
    ward: 'B2',
    admission_date: '2023-10-05'
  },
  {
    _id: ObjectId('6a1ea252e5f996e53a3682d3'),
    patient_id: 'P003',
    name: 'Anjali Deva',
    age: 28,
    gender: 'Female',
    disease: 'Dengue',
    ward: 'B1',
    admission_date: '2023-11-01'
  },
  {
    _id: ObjectId('6a1ea252e5f996e53a3682d4'),
    patient_id: 'P004',
    name: 'Kumara Silva',
    age: 56,
    gender: 'Male',
    disease: 'Heart Disease',
    ward: 'C1',
    admission_date: '2023-08-15',
    status: 'Critical'
  },
  {

```